



Multi-layer Intelligent Evaluation for Smart Contracts

Smart Contract Security Audit

/contracts

Client

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1. Executive Summary

1.1 Key Takeaways

In this Smart Contract Audit Report for the contracts under review, we identified a total of 15 issues, with the majority being low-risk concerns. However, there are three medium-risk findings that warrant immediate attention due to their potential impact on the system's security and overall functionality.

The most significant risk pertains to incorrect equality checks within the contract code. This issue could potentially lead to unintended transactions or misappropriation of funds if not addressed promptly. The financial implications of such a vulnerability could be substantial, as it might allow an attacker to manipulate the system and exploit its weaknesses.

Additionally, we found several low-risk issues, including missing zero checks and timestamp inconsistencies. While these issues do not pose an immediate threat, they can contribute to potential errors or inefficiencies within the contract's operations.

In light of these findings, it is recommended that a thorough review and correction of the identified issues be undertaken before deploying the smart contracts. Ensuring the security and integrity of your system will help mitigate financial risks and maintain user trust, which are crucial for the long-term success of your project.

We stand ready to assist in resolving these issues and improving the overall security posture of your smart contracts. By addressing these concerns, we can help minimize potential business disruptions and ensure a more secure and reliable deployment.

1.2 Deployment Recommendation

Recommendation: CONDITIONAL

Although there are no critical vulnerabilities, the presence of multiple medium and low severity issues may potentially compromise the system's security over time. It is recommended to address these findings before deployment.

1.3 Risk Summary

METRIC	VALUE
Overall Risk Score	48/100
Exploitability	Medium
Business Impact	Medium
Confidence Level	High

Findings by Severity

SEVERITY	COUNT	% OF TOTAL
Critical	0	0.0%
High	0	0.0%
Medium	3	20.0%
Low	6	40.0%
Informational	6	40.0%
Total	15	100%

1.4 Estimated Value at Risk

Value at risk estimation requires additional context about contract TVL and usage patterns.

2. Scope & Methodology

2.1 Engagement Details

PROPERTY	VALUE
Client	Client
Contract	/contracts
Repository	Local Analysis
Commit Hash	N/A
Network	Ethereum Mainnet
Engagement Type	Security Audit

2.2 Scope

In Scope

FILE	LINES	DESCRIPTION
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Total: 1 files, N/A lines of code

Out of Scope

- External dependencies and imported libraries
- Off-chain components
- Economic/tokenomics analysis
- Frontend/backend applications

2.3 Methodology

This audit employed MIESC's comprehensive 9-layer defense-in-depth methodology:

Layer 1: Static Analysis	[--]
Layer 2: Dynamic Testing	[--]
Layer 3: Symbolic Execution	[--]
Layer 4: Formal Verification	[--]
Layer 5: Property Testing	[--]
Layer 6: AI/LLM Analysis	[--]
Layer 7: Pattern Recognition	[--]
Layer 8: DeFi Security	[--]
Layer 9: Advanced Detection	[--]

Tools Utilized

LAYER	TOOL	VERSION	STATUS
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Audit Process

- 1. **Initial Assessment** - Review documentation, understand architecture
- 2. **Automated Analysis** - Execute multi-layer tool suite
- 3. **Manual Review** - Deep dive into flagged code sections
- 4. **AI Correlation** - Cross-reference findings, reduce false positives
- 5. **Verification** - Reproduce and validate vulnerabilities
- 6. **Report Generation** - Document findings with remediation guidance

2.4 Limitations

- Time-boxed engagement (N/A)
- Analysis based on code snapshot at commit N/A
- No guarantee of finding all vulnerabilities
- Economic attack vectors not fully modeled
- Dependency vulnerabilities may exist beyond analysis scope

3. Risk Assessment

3.1 Risk Matrix

The following matrix maps findings by **Impact** (vertical) and **Likelihood** (horizontal):

		LIKELIHOOD		
		Low	Medium	High
IMPACT →	High	Medium 0	High 0	Critical 0
	Med	Low 0	Medium 3	High 0
	Low	Info 6	Low 6	Medium 0

3.2 CVSS-like Scoring

FINDING ID	TITLE	BASE SCORE	VECTOR
slither-incorrect-equality-0	incorrect-equality	4.3	AV:N/AC:L/PR:L/UI:N/C:N/I:L/A:N
slither-missing-zero-check-1	missing-zero-check	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-timestamp-2	timestamp	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-assembly-3	assembly	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-assembly-4	assembly	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-incorrect-equality-0	incorrect-equality	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-missing-zero-check-1	missing-zero-check	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-timestamp-2	timestamp	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-assembly-3	assembly	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-assembly-4	assembly	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-incorrect-equality-0	incorrect-equality	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-missing-zero-check-1	missing-zero-check	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-timestamp-2	timestamp	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-assembly-3	assembly	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N
slither-assembly-4	assembly	3.1	AV:N/AC:H/PR:L/UI:N/C:N/I:L/A:N

Scoring Methodology: - **Attack Vector (AV):** Network, Adjacent, Local, Physical - **Attack Complexity (AC):** Low, High - **Privileges Required (PR):** None, Low, High - **User Interaction (UI):** None, Required - **Impact:** Confidentiality, Integrity, Availability

3.3 Risk Narrative

The audit results indicate several potential security vulnerabilities in the contract, with a focus on assembly code, incorrect equality checks, missing zero checks, and timestamp handling. These findings pose moderate risks to the contract's security.

Primary attack vectors include unintended function calls or data manipulation due to improper assembly code handling (assembly issues), potential bypasses of intended logic due to incorrect equality checks, arithmetic overflows or underflows due to missing zero checks, and tampering with timestamps for malicious purposes (timestamp issues).

The likelihood of exploitation varies depending on the specific implementation details. However, given the nature of these vulnerabilities, an attacker with a moderate level of technical expertise could potentially leverage them to compromise the contract's security.

Immediate actions should include: 1. Reviewing and correcting assembly code to ensure intended functionality and prevent unintended function calls or data manipulation. 2. Implementing proper equality checks to maintain intended logic and protect against arithmetic overflows/underflows. 3. Adding zero checks to prevent potential arithmetic issues related to zero values. 4. Ensuring timestamp handling is secure and resistant to tampering.

Addressing these issues will help strengthen the contract's security posture and reduce the risk of successful attacks. It is recommended to prioritize these tasks based on their impact and complexity, and to conduct regular security audits moving forward.

4. Findings Overview

4.1 Summary Table

ID	TITLE	SEVERITY	STATUS	CVSS
F-001	incorrect-equality	MEDIUM	open	4.3
F-002	missing-zero-check	LOW	open	3.1
F-003	timestamp	LOW	open	3.1
F-004	assembly	INFO	open	3.1
F-005	assembly	INFO	open	3.1
F-006	incorrect-equality	MEDIUM	open	4.3
F-007	missing-zero-check	LOW	open	3.1
F-008	timestamp	LOW	open	3.1
F-009	assembly	INFO	open	3.1
F-010	assembly	INFO	open	3.1
F-011	incorrect-equality	MEDIUM	open	4.3
F-012	missing-zero-check	LOW	open	3.1
F-013	timestamp	LOW	open	3.1
F-014	assembly	INFO	open	3.1
F-015	assembly	INFO	open	3.1

4.2 Category Distribution

CATEGORY	COUNT	SEVERITY BREAKDOWN
assembly	6	6 info
incorrect-equality	3	3 medium
missing-zero-check	3	3 low
timestamp	3	3 low

4.3 Layer Coverage Analysis

LAYER	TOOLS RUN	PASSED	FAILED	FINDINGS	COVERAGE

5. Detailed Findings

F-001. incorrect-equality

PROPERTY	VALUE
Severity	MEDIUM
Category	incorrect-equality
CVSS Score	4.3
Location	ModelRegistryV2.sol:154 (modelExists)
Status	open
Detected By	unknown

Description

ModelRegistryV2.modelExists(bytes32) uses a dangerous strict equality: - models[modelId].createdAt == 0

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Limited financial impact or requires specific conditions to exploit.

Recommendation

Review and fix the vulnerability

References

- No references available

F-002. missing-zero-check

PROPERTY	VALUE
Severity	LOW
Category	missing-zero-check
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/access/Ownable2Step.sol:43 (newOwner)
Status	open
Detected By	unknown

Description

Ownable2Step.transferOwnership(address).newOwner lacks a zero-check on : -
_pendingOwner = newOwner

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Minor impact on contract functionality or gas efficiency.

Recommendation

Review and fix the vulnerability

References

- No references available

F-003. timestamp

PROPERTY	VALUE
Severity	LOW
Category	SC08: Bad Randomness / Front-Running
CVSS Score	3.1
Location	ModelRegistryV2.sol:491 (isAuthorized)
Status	open
Detected By	unknown

Description

ModelRegistryV2.isAuthorized(bytes32,address) uses timestamp for comparisons Dangerous comparisons: - models[modelId].owner == addr || authorized[modelId][addr]

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Minor impact on contract functionality or gas efficiency.

Recommendation

Review and fix the vulnerability

References

- No references available

F-004. assembly

PROPERTY	VALUE
Severity	INFO
Category	assembly
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/utils/StorageSlot.sol:66 (getAddressSlot)
Status	open
Detected By	unknown

Description

StorageSlot.getAddressSlot(bytes32) uses assembly - INLINE ASM

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Informational finding for code quality improvement.

Recommendation

Review and fix the vulnerability

References

- No references available

F-005. assembly

PROPERTY	VALUE
Severity	INFO
Category	assembly
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/utils/StorageSlot.sol:102 (getInt256Slot)
Status	open
Detected By	unknown

Description

StorageSlot.getInt256Slot(bytes32) uses assembly - INLINE ASM

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Informational finding for code quality improvement.

Recommendation

Review and fix the vulnerability

References

- No references available

F-006. incorrect-equality

PROPERTY	VALUE
Severity	MEDIUM
Category	incorrect-equality
CVSS Score	4.3
Location	ModelRegistryV2.sol:154 (modelExists)
Status	open
Detected By	unknown

Description

ModelRegistryV2.modelExists(bytes32) uses a dangerous strict equality: - models[modelId].createdAt == 0

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Limited financial impact or requires specific conditions to exploit.

Recommendation

Review and fix the vulnerability

References

- No references available

F-007. missing-zero-check

PROPERTY	VALUE
Severity	LOW
Category	missing-zero-check
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/access/Ownable2Step.sol:43 (newOwner)
Status	open
Detected By	unknown

Description

Ownable2Step.transferOwnership(address).newOwner lacks a zero-check on : -
_pendingOwner = newOwner

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Minor impact on contract functionality or gas efficiency.

Recommendation

Review and fix the vulnerability

References

- No references available

F-008. timestamp

PROPERTY	VALUE
Severity	LOW
Category	SC08: Bad Randomness / Front-Running
CVSS Score	3.1
Location	ModelRegistryV2.sol:491 (isAuthorized)
Status	open
Detected By	unknown

Description

ModelRegistryV2.isAuthorized(bytes32,address) uses timestamp for comparisons Dangerous comparisons: - models[modelId].owner == addr || authorized[modelId][addr]

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Minor impact on contract functionality or gas efficiency.

Recommendation

Review and fix the vulnerability

References

- No references available

F-009. assembly

PROPERTY	VALUE
Severity	INFO
Category	assembly
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/utils/StorageSlot.sol:66 (getAddressSlot)
Status	open
Detected By	unknown

Description

StorageSlot.getAddressSlot(bytes32) uses assembly - INLINE ASM

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Informational finding for code quality improvement.

Recommendation

Review and fix the vulnerability

References

- No references available

F-010. assembly

PROPERTY	VALUE
Severity	INFO
Category	assembly
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/utils/StorageSlot.sol:102 (getInt256Slot)
Status	open
Detected By	unknown

Description

StorageSlot.getInt256Slot(bytes32) uses assembly - INLINE ASM

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Informational finding for code quality improvement.

Recommendation

Review and fix the vulnerability

References

- No references available

F-011. incorrect-equality

PROPERTY	VALUE
Severity	MEDIUM
Category	incorrect-equality
CVSS Score	4.3
Location	ModelRegistryV2.sol:154 (modelExists)
Status	open
Detected By	unknown

Description

ModelRegistryV2.modelExists(bytes32) uses a dangerous strict equality: - models[modelId].createdAt == 0

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Limited financial impact or requires specific conditions to exploit.

Recommendation

Review and fix the vulnerability

References

- No references available

F-012. missing-zero-check

PROPERTY	VALUE
Severity	LOW
Category	missing-zero-check
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/access/Ownable2Step.sol:43 (newOwner)
Status	open
Detected By	unknown

Description

Ownable2Step.transferOwnership(address).newOwner lacks a zero-check on : -
_pendingOwner = newOwner

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Minor impact on contract functionality or gas efficiency.

Recommendation

Review and fix the vulnerability

References

- No references available

F-013. timestamp

PROPERTY	VALUE
Severity	LOW
Category	SC08: Bad Randomness / Front-Running
CVSS Score	3.1
Location	ModelRegistryV2.sol:491 (isAuthorized)
Status	open
Detected By	unknown

Description

ModelRegistryV2.isAuthorized(bytes32,address) uses timestamp for comparisons Dangerous comparisons: - models[modelId].owner == addr || authorized[modelId][addr]

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Minor impact on contract functionality or gas efficiency.

Recommendation

Review and fix the vulnerability

References

- No references available

F-014. assembly

PROPERTY	VALUE
Severity	INFO
Category	assembly
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/utils/StorageSlot.sol:66 (getAddressSlot)
Status	open
Detected By	unknown

Description

StorageSlot.getAddressSlot(bytes32) uses assembly - INLINE ASM

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Informational finding for code quality improvement.

Recommendation

Review and fix the vulnerability

References

- No references available

F-015. assembly

PROPERTY	VALUE
Severity	INFO
Category	assembly
CVSS Score	3.1
Location	lib/openzeppelin-contracts/contracts/utils/StorageSlot.sol:102 (getInt256Slot)
Status	open
Detected By	unknown

Description

StorageSlot.getInt256Slot(bytes32) uses assembly - INLINE ASM

Vulnerable Code

```
// No code snippet
```

Impact Analysis

Informational finding for code quality improvement.

Recommendation

Review and fix the vulnerability

References

- No references available

6. Remediation Roadmap

6.1 Prioritized Actions

PRIORITY	FINDING	SEVERITY	EFFORT	RATIONALE
1	incorrect-equality	Medium	Medium	Incorrect equality vulnerabilities can lead to unintended contract behavior and potential exploitation. The issue is found twice in ModelRegistryV2.
2	timestamp	Low	Medium	Missing zero-checks can potentially allow transactions with zero values, which might lead to unintended contract behavior or exploitation. The issue is found twice in ModelRegistryV2 and Ownable2Step.
3	incorrect-equality	Medium	Medium	Repeated incorrect equality vulnerabilities can lead to unintended contract behavior and potential exploitation. The issue is found twice in ModelRegistryV2.
4	missing-zero-check	Low	Medium	Missing zero-checks can potentially allow transactions with zero values, which might lead to unintended contract behavior or exploitation. The issue is found twice in Ownable2Step.
5	assembly	Info	Medium	Assembly usage may not be harmful but it's good practice to review assembly code for potential vulnerabilities.
6	assembly	Info	Medium	Assembly usage may not be harmful but it's good practice to review assembly code for potential vulnerabilities.
7	timestamp	Low	Medium	Timestamp dependency check might not pose a significant risk, but it's good practice to review such checks for potential vulnerabilities.
8	assembly	Info	Medium	Assembly usage may not be harmful but it's good practice to review assembly code for potential vulnerabilities.
9	assembly	Info	Medium	Timestamp dependency check might not pose a significant risk, but it's good practice to review such checks for potential vulnerabilities.

6.2 Remediation Timeline

PHASE	WEEK	PRIORITY	FINDINGS	ACTION
1	Week 1	Critical & High	0 + 0	Immediate remediation required
2	Week 2	Medium	3	Address medium severity issues
3	Week 3	Low & Info	6 + 6	Fix low priority items
4	Week 4	Verification	-	Re-audit and validation

6.3 Quick Wins

Review findings marked as "Low" effort for quick security improvements.

6.4 Effort vs Impact Matrix

		IMPACT		
		Low	Medium	High
	High	Avoid	Consider	Schedule
	Med	Defer	Plan	Priority
	Low	If Time	Quick Win	DO FIRST!
		EFFORT →		

7. Appendices

Appendix A: Tool Execution Details

Appendix B: Files Analyzed

#	FILE PATH	LINES	FUNCTIONS	FINDINGS
1	/contracts	N/A	--	15

Appendix C: SWC Registry Compliance

SWC ID	TITLE	STATUS	FINDING(S)
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Appendix D: OWASP Smart Contract Top 10

RANK	CATEGORY	STATUS	FINDINGS
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Appendix E: Glossary

TERM	DEFINITION
Reentrancy	A vulnerability where an external call allows execution to re-enter the calling contract before the first execution completes
Integer Overflow	When an arithmetic operation results in a value larger than can be stored in the variable
Front-running	Exploiting knowledge of pending transactions to gain an advantage
Flash Loan Attack	Using uncollateralized loans within a single transaction to manipulate protocols
Oracle Manipulation	Attacking price feeds or external data sources to influence contract behavior
Access Control	Mechanisms that restrict who can execute sensitive functions

Appendix F: Audit Trail

Event	Timestamp	Hash
Contract Snapshot	2026-01-24	N/A
Analysis Started	2026-01-24	--
Analysis Completed	2026-01-24 21:33:35	--
Report Generated	2026-01-24 21:33:35	--
Report Hash	2026-01-24 21:33:35	N/A

Disclaimer

This audit report is provided on an "AS IS" basis without warranties of any kind, whether express or implied. The findings and recommendations represent the auditor's professional opinion based on the scope and methodology described.

Limitations: - This audit does not guarantee the absence of vulnerabilities - Smart contract security is an evolving field - New vulnerabilities may be discovered post-audit - Economic and governance attacks may not be fully modeled - The client is responsible for implementing and testing fixes

AI Disclosure: Sections labeled as "AI-Generated" or "AI Analysis" were produced using large language models (mistral:latest). These AI-generated insights are supplementary and should be reviewed by qualified security professionals. AI outputs may contain inaccuracies and are provided for informational purposes only.

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